AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

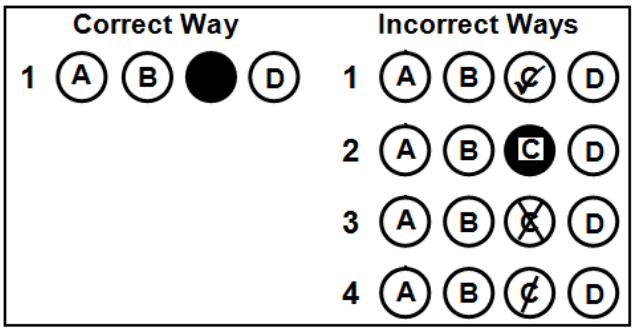
Biology Paper I

Time: 35 minutes Marks: 25

AKU-EB May 2018 for Teaching & Learning Only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. The structures of the respiratory system of human beings between which exchange of carbon dioxide and oxygen gas takes place are

   A. alveoli and capillaries.
   B. trachea and capillaries.
   C. alveoli and bronchioles.
   D. trachea and bronchioles.

2. Breathlessness is one of the major symptoms of emphysema. Breathlessness occurs because the

   A. patient suffers from cough.
   B. air spaces of alveoli are enlarged.
   C. walls of bronchioles become thin.
   D. heart cannot pump enough oxygenated blood.

3. Respiration is different from gaseous exchange because respiration

   A. consists of a series of chemical reactions.
   B. requires a concentration gradient to occur.
   C. involves lungs and its associated structures.
   D. takes place in higher multicellular organisms.

4.

- Sunken stomata
- Succulent organs
- Thick waxy cuticle

A plant with the given characteristics would belong to

   A. deserts.
   B. mountains.
   C. fresh water.
   D. marine water.

5. The similarity between the working of dialysis machine and human kidneys is that both

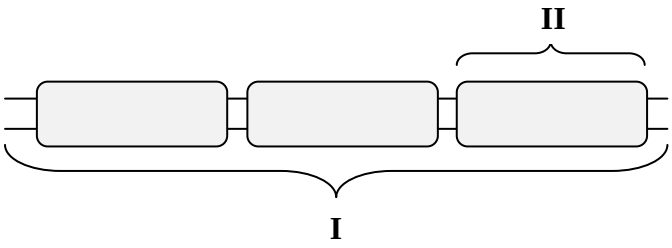
   A. reabsorb proteins into blood.
   B. remove macromolecules from blood.
   C. maintain the mineral balance in blood.
   D. change oxygenated blood to deoxygenated blood.

6. Which of the following is CORRECT about the gland and the hormone it secretes?

	Gland	**Hormone**
A	Thyroid gland	Thyroid stimulating hormone
B	Pituitary	Vasopressin
C	Parathyroid	Calcitonin
D	Ovary	Oxytocin

AKU-EB May 2018 for Teaching & Learning Only

7. The given diagram shows a part of a neuron.



The labelled structures, **I** and **II** are

	I	**II**
A	Axon	Schwan cell
B	Axon	Node of Ranvier
C	Dendrite	Schwan cell
D	Dendrite	Node of Ranvier

8. When Zain encounters a fighter bull, he experiences a sudden increase in his heart beat. The increase in heart beat would

A. increase the rate of digestion.
B. decrease the air flow to his lungs.
C. convert maximum glucose to glycogen.
D. transfer glucose and oxygen to his muscles.

9. Which of the following features CORRECTLY differentiates between bone and cartilage?

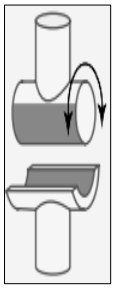
	Bone	**Cartilage**
A	Composed of chondrocytes	Composed of osteocytes
B	Cannot grow	Can grow
C	Blood vessels penetrate	Blood vessels do not penetrate
D	Nerves do not penetrate	Nerves penetrate

AHSEC-EB May 2018 for Teaching & Learning Only

10. The parts of human body containing **joint I** and **joint II** are



Joint I



Joint II

	Body Part Containing Joint I	**Body Part Containing Joint II**
A	Elbow	Knee
B	Shoulder	Elbow
C	Knee	Hip
D	Hip	Shoulder

11. Osteoporosis is a bone disorder in which the density of bones decreases.

The prevalence of this disease in elderly women is MOST likely related to the

A. deficiency of estrogen.
B. deficiency of parathormone.
C. excessive consumption of vitamin C.
D. excessive deposition of mineral salts.

12. Which of the following is CORRECT about the number of chromosome sets present in the structures involved in the life cycle of a flowering plant?

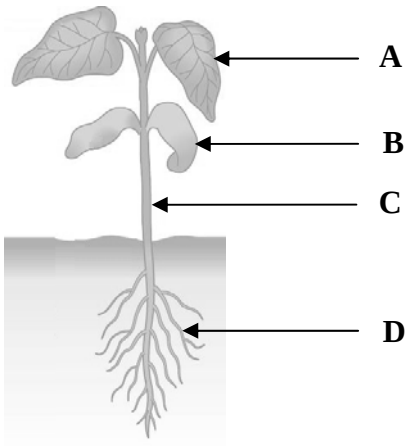
	Structure	**Chromosome Set**
A	Macrospore	2n
B	Generative nucleus	n
C	Endosperm nucleus	2n
D	Microspore mother cell	n

13. A flower is MOST likely pollinated by wind if it has

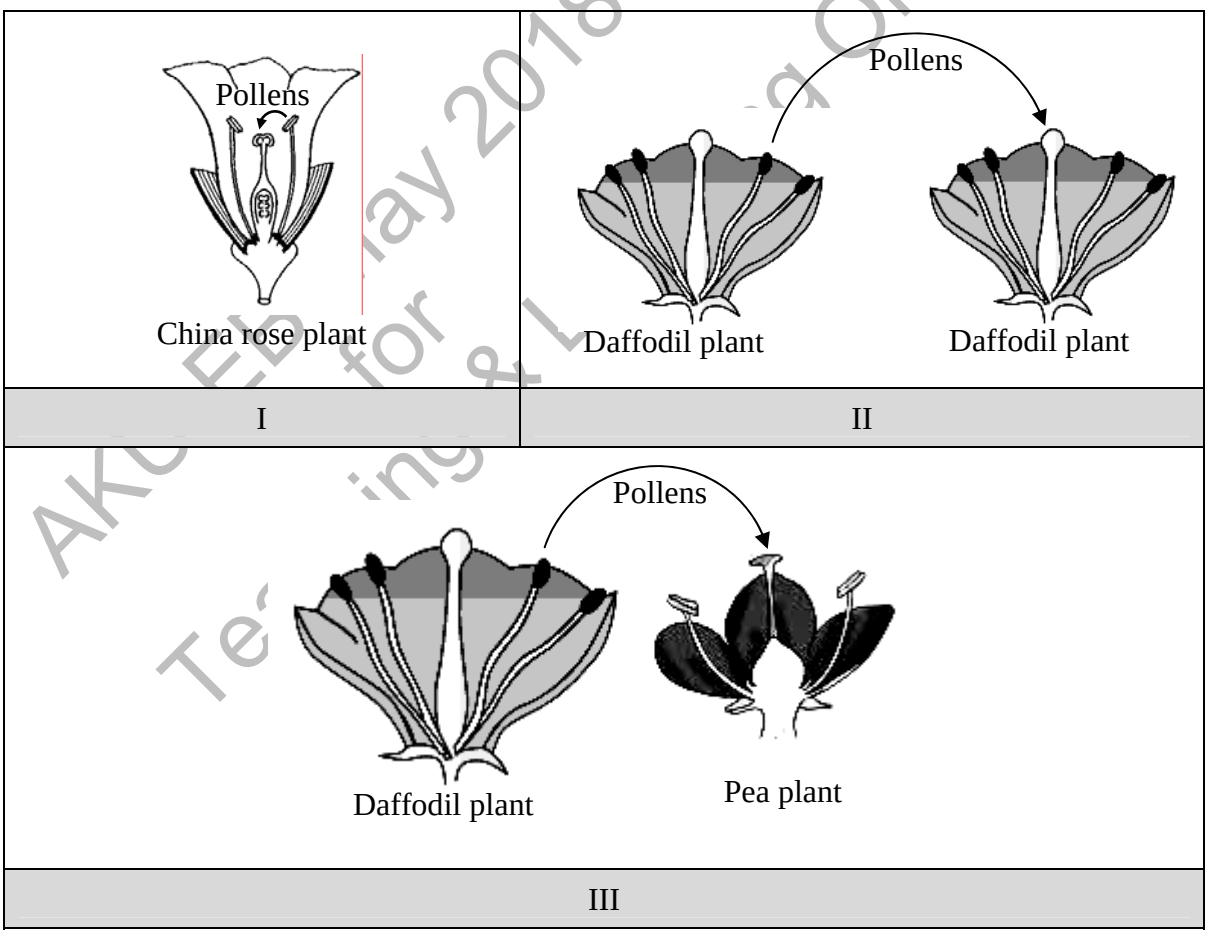
A. small pink petals with nectar.
B. large red petals with feathery stigmas.
C. large white petals with sticky pollen grains.
D. small white petals with long slender filaments.

AKU-EB May 2018 for Teaching & Learning Only

14. The structure of a seedling that provides food to the developing embryo during the process of germination is



15. Fertilisation will occur in which of the following cases?



A. II only
B. III only
C. I and II
D. I and III

AKU EB May 2018 for Teaching & Learning Only

16. An example of co-dominance is

	Parent Phenotype	Offspring Phenotype
A	A and O blood group	A blood group
B	Black and white fur	Black fur with white stripes
C	Round and wrinkled seed	Round seed
D	Red and white flower	Pink flower

17. The number of different types of gametes formed from **QqRr** would be

A. 1
B. 2
C. 4
D. 8

18. While conducting an experiment, a scientist crosses a red flower with a white flower and finds 100% pink flowers.

The MOST relevant conclusion of the given cross would be

A. red parent flower is heterozygous.
B. red colour is co-dominant with white.
C. pink offspring flowers are homozygous.
D. white colour carries both recessive alleles.

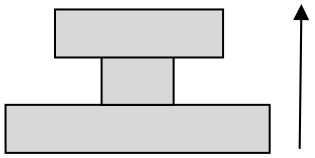
19. The pyramid of number that CORRECTLY represents the given food chain is

Food Chain: Tree → Birds → Mosquitos

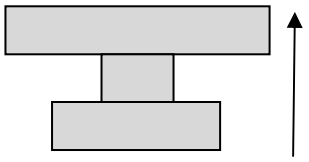


A

B



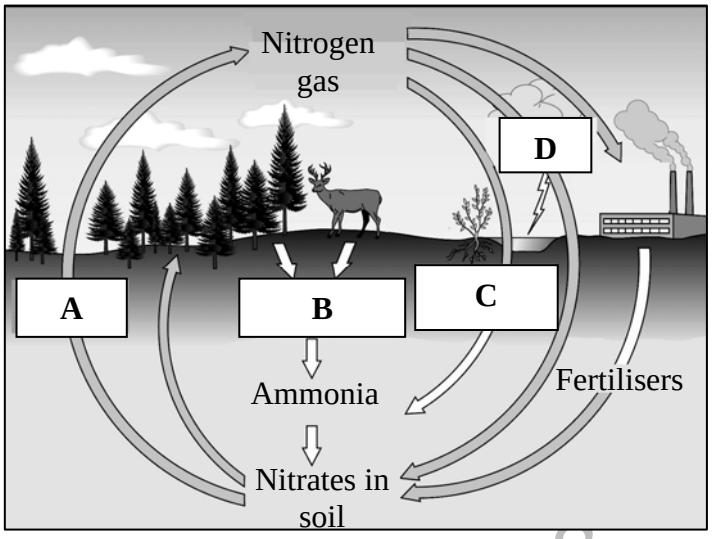
C



D

AKU-EB May 2018 for Teaching & Learning Only

20. Which of the following events of nitrogen cycle decreases soil fertility by depleting nitrogen from it?



21. A species is a group of organisms which

A. interbreeds freely in nature.
B. lives without abiotic factors.
C. consists of unicellular organisms only.
D. contains different number of chromosomes.

22. An application of fermentation is the

A. synthesis of human growth hormone.
B. formation of ethanol using *Saccharomyces*.
C. improvement in the production of enzymes.
D. production of a particular protein molecule.

23. Zara works in a biotechnological laboratory and is interested in the production of human growth hormone (HGH).

The first step Zara would take to start this procedure is the

A. isolaton of HGH gene.
B. insertion of HGH gene in plasmid.
C. transfer of recombinant DNA in bacteria.
D. growth of genetically modified organism.

AKU-EB May 2018 for Teaching & Learning Only

24. Drug **X** has the following characteristics.

I. Relieves pain
II. Acts directly on central nervous system
III. High potential for addiction

Drug **X** is identified as

A. diazepam.
B. morphine.
C. tetracycline.
D. polio vaccine.

25. A person suffering from nervousness and insomnia (sleeplessness) should be treated by prescribing

A. sedatives.
B. antiseptics.
C. pain killers.
D. hallucinogens.

AKU-EB May 2018 for Teaching & Learning Only

END OF PAPER

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only